Coupling Coarse Earth Models(CCEM), Global Warming and Social Cost of Carbon

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November 2024

2025 Model 8 Formal Presentation

Latest update : January 3rd, 2026

Energy Resource Model (M1)

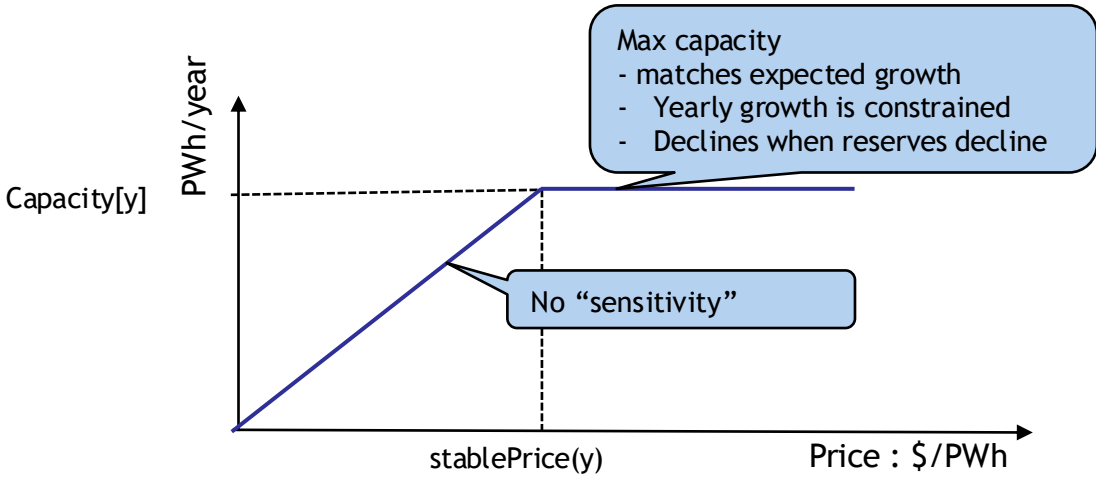
M1 captures the answer to the questions « ***How much fossil resources do we have ? At which costs?*** »

Four categories

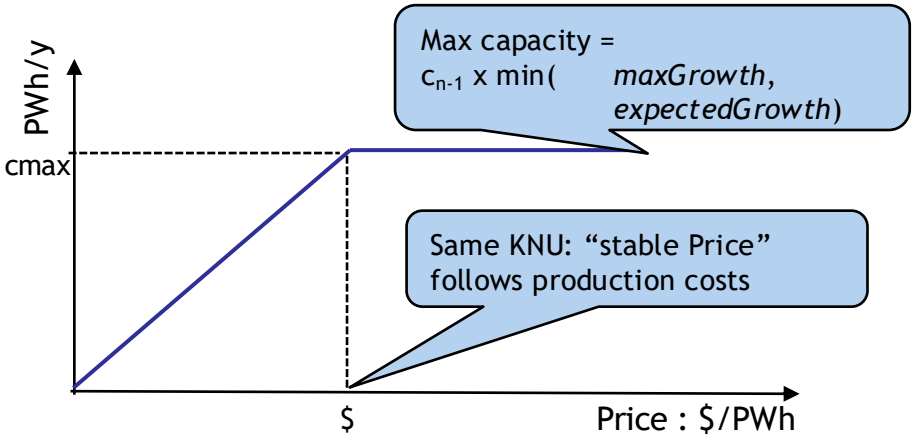
- Oil
- Natural Gas
- Coal
- Clean (hydro/solar/wind/nuclear)

Each resource is defined through:

- For fossil energies, its *inventory chart* (available Toe according to sell price)
- For clean energies, a *growth potential* curve (max capacity in the future year, according to manufacturing and resource constraints)
- A “max capacity” (max yearly output)
- A constraint about the speed at which this capacity may evolve + a heuristic formula that adjusts the capacity each year
- An ”stable price” (KNU: affine as function of time) that tells the price when there are no shortages nor surpluses



Fossil Fuel Production = f(price)



Clean Energy Production = f(price)

M1: Energy Production

Belief: Known Unknowns

Data-Driven Parameters

Data Model

Two kinds of sources of primary energy (e):

- FossilFuels :Oil, Goal, Gas
- One combined "Clean" (nuclear & renewable)

Energy sources are described through the max capacity (that varies), the yearly output (depending on supply-demand) and, for fossil fuels, the known reserves

The main component of M1 is the Supply equation that describes output as a function of price

Parameters

- *maxCapacityGrowth*(e,y) : for clean energy y, expected max caoacity in Gtoe that may be added during year y (yearly production)
- *Inventory*(e,p) : expected reserves (at year 1) for fossil fuel e with a market price p
- *threshold*(e) : part of current reserve when suppliers of e reduce their output to match reserves (strong influence on PeakOil date)
- stablePrice(e,y): expected price (constant dollars) of energy *e* in year y, assuming no shortages nor surpluses

- *targetMaxRatio*(e) : expected ratio between (max) capacity and output
- *maxGrowthRate*(e): percentage of capacity that can be added at most in a year for energy e
- *co2perTon(e)* : CO2 emissions to produce one toe of energy e

State Variables:

- $O_e(y)$: output (production) in PWh for energy e at year y
- $C_e(y)$: max capacity in PWh for energy e
- $A_e(y)$: added capacity for energy e through transfers (M3)
- $tO_e(y)$: total output in PWh from years 1 to y
- $P_e(y)$: price in $ for 1 toe for energy e at year y
- $UD_z(y)$: demand (unconstrained consumption) for zone z of energy e
- $G_z(y)$: gdp for zone z on year y

Equations (functions & differential state)

[1][2] $Supply(e{:}Energy,p,y,C_{max}) = \min(C_{max}, \max(0, O_e(1) \times \min(1,(C_{max} / C_e(1))) \times P_e(1) / stablePrice(e,y))$

[3] $Capacity(e{:}Fossil,y) = InventoryToCapacity(e, C_e(y-1) \times (1 + \min(expectedGrowth(e,y), maxGrowthRate(e))), (Inventory(e,p) - tO_e(y-1)) / threshold(e)))$

$InventoryToCapacity(e{:}Fossil,c,cMax,r) =$
if $(r > 1)$ c
else if $(r < 0.5)$ $\min(c, r \times C_e(1) \times 1.75)$
else $\min(c, C_e(1) \times (1 - (1-r)^2 * 0.5)$

[4] $Capacity(e{:}Green,y) = C_e(y-1) + \min(C_e(y-1) \times expectedGrowth(e,y), C_e(y-1) + (maxCapacityGrowth(e,y) - A_e(y-1)),$

[5] $expectedGrowth(e,y) = \max(0, [\Sigma_{z \in zone}\, linearRegression(UD_z(y-3), UD_z(y-2)\, UD_z(y-1))] / C_e(y-1) - 1)$

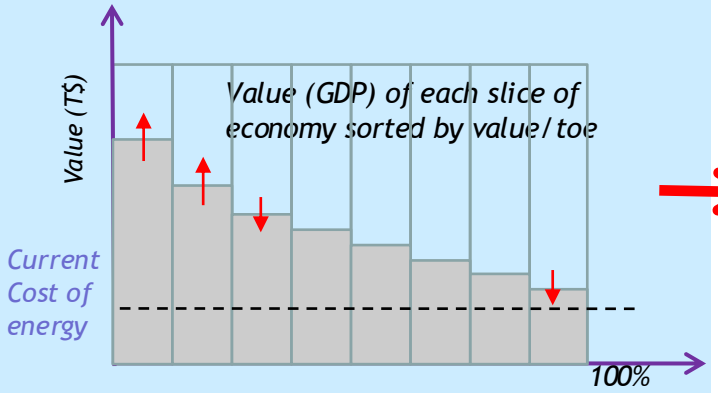
Energy Consumption Model (M2)

M2 captures the answer to the question

« ***How much energy is needed for the economy at a given cost ?***»

The heart of this model is (for each source & each zone) a histogram of value production:

- Decomposition of value product (Y axis)
- Over "segments" of energy usage (X axis), sorted by energy intensity
- This (virtual) decomposition is the base for telling how each world zone will react to price increases



Evolution of distribution represents the dematerialization of the zone economy (kW/$)

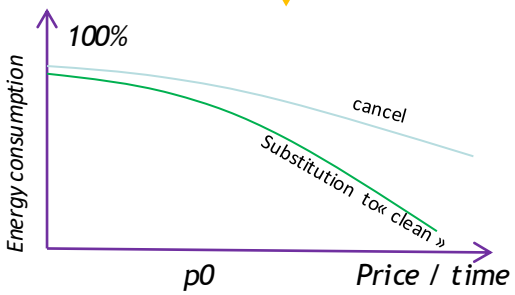
The net behavior is represented by four curves :

- Economy dematerialization trend (kW/$)
- Cancelation (% of activity that stops because price is too high)
- Economic loss due to cancelation (GDP impact of cancel)
- Margin reduction

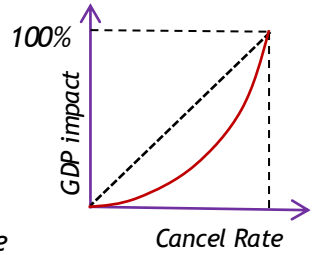
In addition, we represent the expected efficiency gains:

- Substitution towards another form of energy (using the matrix provided by M3)
- The last two options triggers associated investments

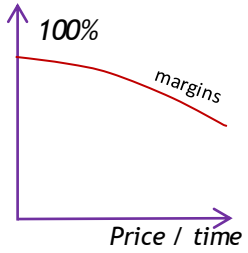
Energy redistribution policy is a factor that lowers the pain of cancellation but reduces the economic efficiency



Energy consumption for a given price decreases according to cancelation (upper histogram), energy savings and substitution (M3: one energy source to another)



The loss of economic value due to Energy-based cancellation follows a convex law (lower value creation activities stop first)



Margins for surviving activities are reduced by increasing energy prices

M2: Energy Consumption

Belief

Data-driven

Data Model

- Four Energy sources, Four zones : EU,US, CN, RoW
- M2 is based on analyzing the gdp production through "homogeneous slices" to state 4 beliefs: Energy density, cancel, GDP & margin impact
- M2 factors efficiency gains through technology through the "saving" parameter for each zone
- M3 described the "energy transition" that transfers some of the needs from one energy source to another

Parameters

- *cancel*(z,p) : share (percentage) of economy for zone z if the oil price equivalent reaches p
- *impact*(z,p) : associated impact on gdp (output of the remaining activities) when price is p
- *dematerialize*(z,y) : expected decline in energy density (gdp/conso) for zone z

- *economyRatio*(z,y) : heuritiscs that combines the expected growth of the zone gdp (from the amount of past investments) and the mutual influence of zones through global trade
- pop2energy(z): ratio between energy consumption growth and population growth
- Population(z,y) : adjusted population forecast that factors birth decline (M5)

State Variables:

- $R_z(e,y)$: raw needs for energy e in Gtoe at year y (before efficiency or transition is applied)
- $N_z(e,y)$: needs for energy e in zone z during year y once energy transition transfers are applied
- $Tr(e_1,e_2,y)$: fraction of energy e_1 demand that has been transferred to energy source e_2 at year y
- $U_z(e,y)$: usage (constrained consumption) for zone z of energy e
- $P_e(y)$: Price for energy e ($/toe) at year y
- $S_z(y)$: percentage of savings reached at year y
- $GW_z(y)$: percentage of capacity lost because of global warming, cumulative to year y

Equations (functions & differential state)

[1] $R_z(e,y) = U_z(e,1) \times \mathit{economyRatio}(z,y) \times \mathit{energyIntensity}(z,y)) \times (1 - GW_z(y-1))$

[2] $\mathit{energyIntensity}(z,y) = (1 - \mathrm{dematerialize}(z,y)) \times (1 - S_z(y))$

[3] $\mathit{economyRatio}(z,y) = (M_z(y-1) / M_z(1)) \times \mathrm{tradeRatio}(z,y)$
$\mathrm{tradeRatio}(z,y) = \mathrm{innerTrade}(z) + \mathrm{outerCommerceRatio}(z,y) + \mathrm{importReductionRatio}(z,y)$
$\mathrm{outerCommerceRatio}(z,y) = \Sigma_{z1 \neq z} (\mathit{trade}(z,z1) \times \mathit{protect}(z,z1) \times \mathrm{protectionismOutFactor})$
$\mathrm{importReductionRatio}(z,y) = \Sigma_{z1 \neq z} (\mathit{trade}(z1,z) \times \mathit{protect}(z1,z) \times (M_z(1) / M_{z1}(1)) \times \mathrm{protectionismInFactor})$

[4] $N_z(e,y) = R_z(e,y) + \Sigma_{e1<e} R_z(e1,y) \times Tr(e1,e,y) - \Sigma_{e<e2} R_z(e2,y) \times Tr(e,e2,y)$

[5] $\mathit{Demand}(e,z,y,p) = N_z(e,y) \times (1 - S_z(y-1)) \times (1 - \mathit{cancel}(z,p + \mathrm{tax}(z,e,y)))$
$\mathit{Demand}(e,y,p) = \Sigma_z \mathrm{Demand}(e,z,y,p)$
$\mathit{tax}(z,e,y) = \mathit{co2Tax}(z,CO2(y-1))$

[6] $P_e(y) = !\, p \mid \mathit{Demand}(e,y,p) = \mathit{Supply}(e,y,p)$

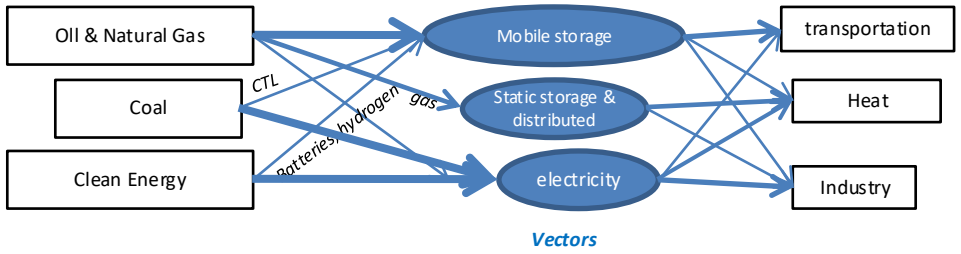
[7] $C_e(y) = \max(C_e(y-1) , \mathit{Capacity}(e,y, P_e(y)))$
$O_e(y) = \Sigma_z \mathit{Supply}(e,z,y,p)$

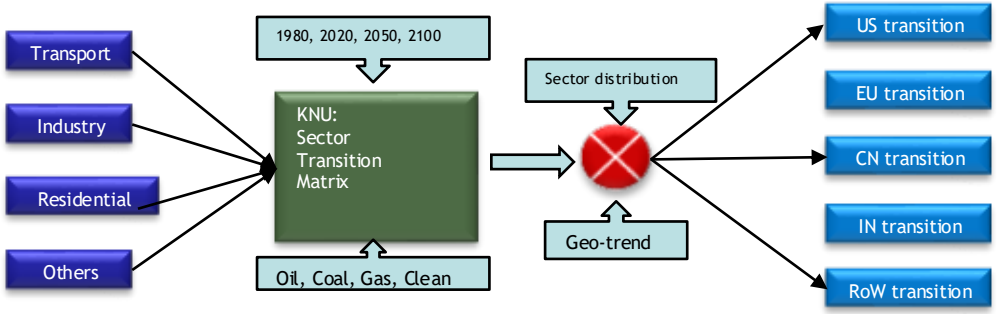
Energy Transition Model (M3)

M3 captures the answer to « ***How fast can we substitute one form of primary energy to another ?*** »

Substitution matrix

- Oil to Coal to Gas to Clean
- 6 flows (N x N-1 / 2)
- A substitution matrix as 6 coefficients (share of energy consumption that should be transitioned), depending on year (policy)
- The matrix takes into account the “source to vector” possible paths (cf. illustration) – It should also factor the constraints from natural resources such as steel.
- The matrices for zones are derived from energy sectors: transport, industry, residential, others

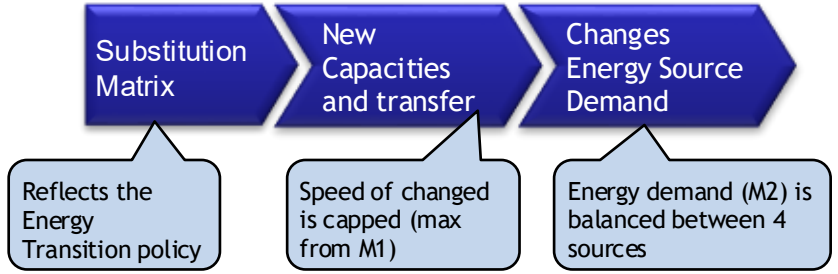




The substitution matrix describes which substitution is feasible & desirable from an energy policy viewpoint (on the demand side)

- Irreversible
- When acted (a given year), generate the associated Investment (same for savings)

Note: *Models M1 / M2 focus on primary energy sources, M3 takes the energy vectors (electricity, hydrogen, …) into account to define which transitions are feasible – this is done for each sector*



M3: Energy Transition

Belief

Data-driven

Data Model

Energy sources are ordered
- Oil < Coal < Gas < Clean
- Yielding 6 transitions (e1 -> e2)

The energy transition is a time-dependant matrix that represent the possible transfer of one primary source another (thanks to industrial investments or the use of vectors such as electricity)

Parameters

- *transitionRate*(z,e_1,e_2, y) : maximum transfer of energy needs from primary source e1 to e2 at year y, expressed as a percent
- *techEfficiency*(z) : yearly growth of tech efficiency (cost reduction in investment capacity)
- investPrice(e) : investment that is necessary to build a capacity of 1Gtoe/y at year 1
- ftech(z) : expected yearly decline of investPrice in zone z (technology progress)
- *steelFactor*(e) : part of steel cost in total cost of investment for e
- *co2Energy*: percentage of CO2 emission due to fossile energy
- *policy(tr,f,y) : applies the factor f to the transition rate when y is more than PivotYear (2020)*
- maxTransferRate(e,y) : share of the total transfers policies to energy e, matching *maxGrowthRate*(e)

State Variables:

- $P_e(y)$: price in $ for 1 toe for energy e at year y
- $U_z(e,y)$: usage (constrained consumption) for zone z of energy e
- $S_z(y)$: percentage of savings reached at year y
- $CN_z(y)$: percentage of consommation canceled in zone z at year y, because the price is too high
- $IE_z(y)$: investments for new energy capacity for energy source z at year y
- SP(y): steel price for year y
- $SvF_z(y)$: acceleration of energy efficiency (saving %)
- $TrF_z(y)$: acceleration of energy transition (factor) this represents a zone policy
- CO2(y): emission for year y in Gt

Equations (functions & differential state)

[1] $U_z(e,y) = N_z(e,y) \times (1 - cancel(z, P_e(y)))$

[2] $S_z(y) = \max(S_z(y), \min(S_z(y) + maxYearlySaving(z), SvF_z(y-1)))$
$CN_z(y) = \Sigma_e\ cancel(z, P_e(y))$

[3] $Tr(e1,e2,y,) = \max(Tr(e1,e2,y-1),$
$\min(policy(transitionRate(z,e1,e2,y), TrF_z(y), y), \max(Tr(e1,e2,y-1) + maxTransferRate(e,y))$

[4] $Tax_z(y) = \Sigma_e\ (O_z(e,y) \times co2perTon(e) \times CarbonTax(z,y)$

[5] $IE_z(y) = \Sigma_e\ [\ (\max(0, C_e(y) - C_e(y-1)) \times (U_z(e,y) / \Sigma_{z1} U_{z1}(e,y)) +$
$(N_z(e,y) \times \max(0, S_z(y) - S_z(y-1))) +$
$\Sigma_{e<e2} N_z(e_2,y) \times \max(0, (Tr(e,e_2,y) - (Tr(e,e_2,y-1)))\ ] \times$
$investPrice(e) \times (1 - steelFactor(e) + (steelFactor(e) * SP(y-1) / SP(1)) \times (1 - ftech)^y$

[6] $Elec_z(y) = \Sigma_e\ (U_z(e,y) \times eRatio_z(e)$
$+ \Sigma_{e1,e2} Tr(e1,e2,y) \times (1 - elec\%(e1)) \times (1 - heat\%(e1,e2)))$

[7] $CO2(y) = (\Sigma_e O_z(e,y) \times co2perTon(e))$

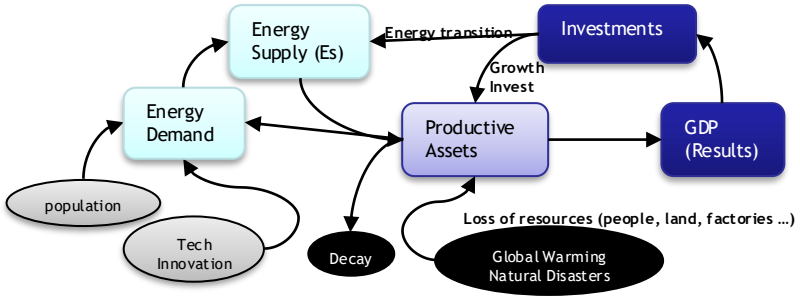
GDP Model (M4, World Economy)

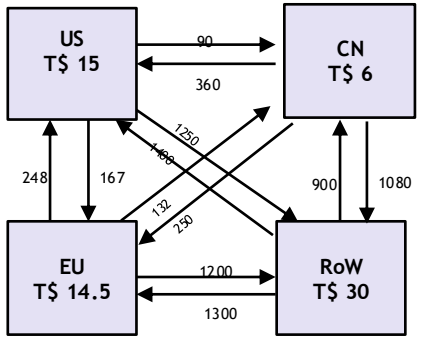
M4 represents the question:
« which GDP is produced from investment, technology, energy and workforce ? »

Description

Economy is seen as a set of productive assets that require energy and human capital to operate

Economy growth require investments that are a fraction of results

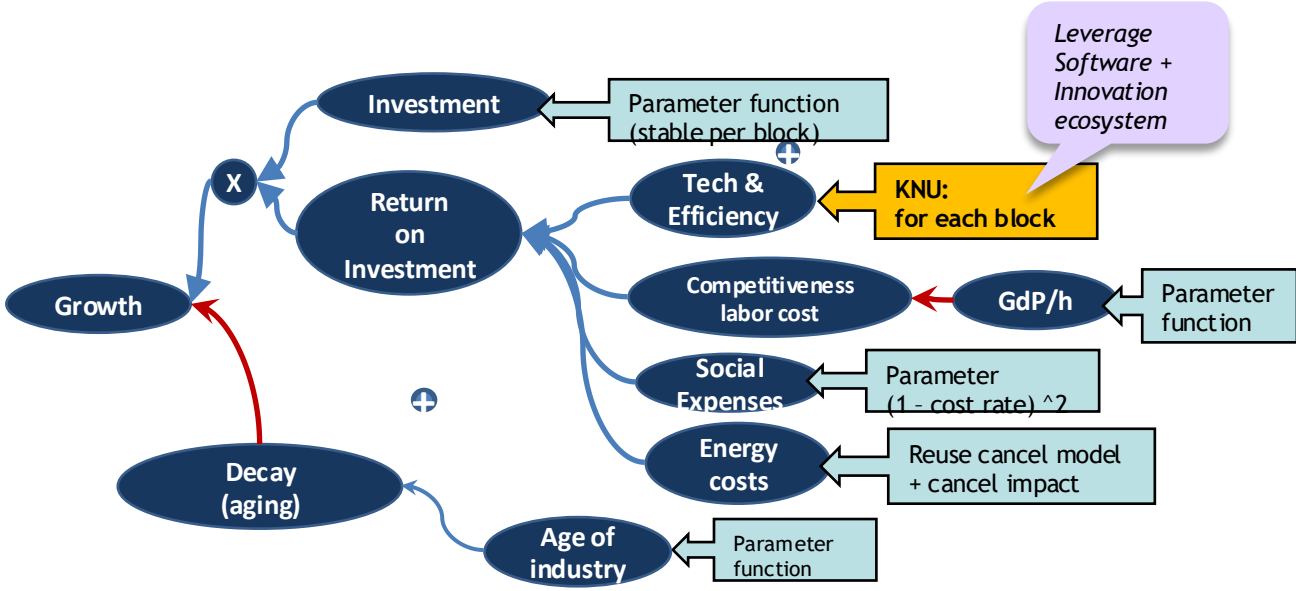




2010 world economy crude decomposition (trade in G$)

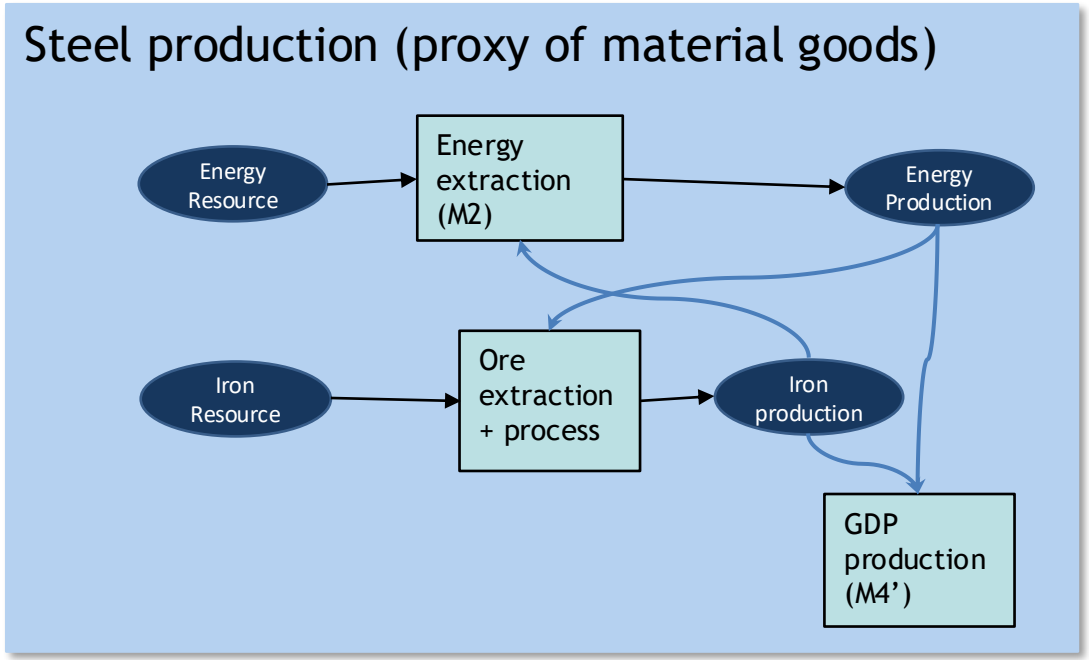
We use a crude exponential growth model

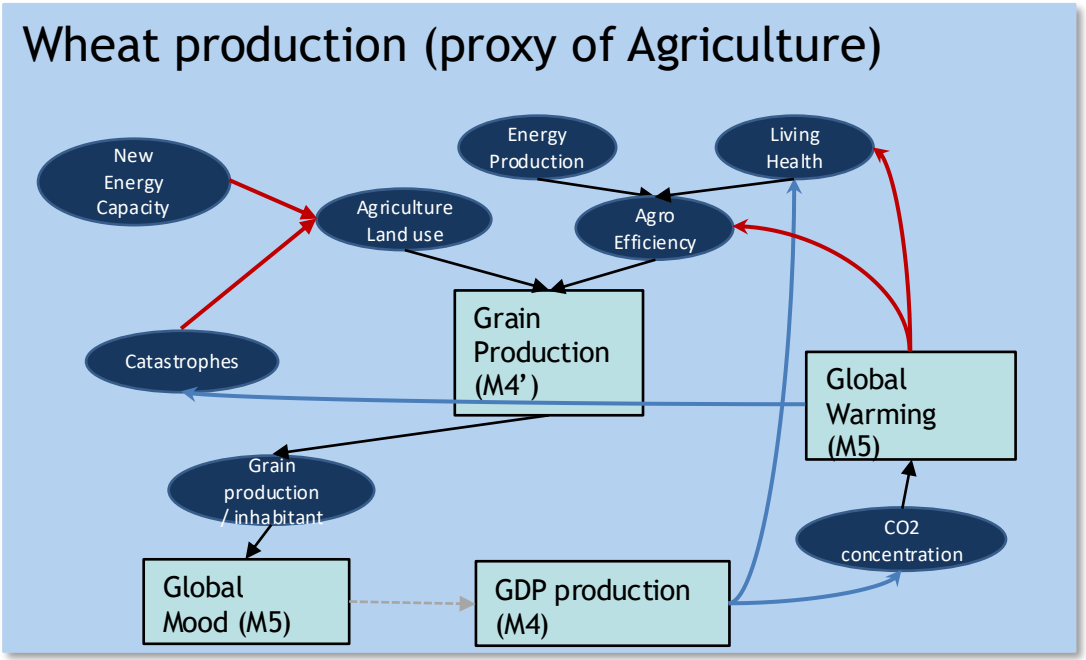
- Output is linked to energy (cf. M2 + redistribution)
  - Demography is factored through energy consumption
- Assets grow as a result of investments – RoI is the combination of multiple factors (social expenses, energy, labor, and a KNU: RoiEfficiency)
- Energy transition investments are subtracted from total investments to compute growth investments
- GW Crises are modeled two ways : a reduction of productive assets and as "redirections of zone policies" (M5)
- The model computes the consumption without savings that drives the GDP and the consumption with savings that drives CO2 emissions
- This model is applied to 5 "Blocks" : US, CN, EU, IN, RoW



M4: Physical versus Immaterial GDP

- The model uses “constant dollars” as monetary unit (for GDP and energy price)
  - 2% inflation used to display “current dollars” results … but ”2100 dollars” is an abstract notion ☺
- High GDP growth in the past 30 years reflects the growing importance of “immaterial” economy
- The “$CCEM_4$” model (4th iteration) adds two outputs to evaluate the “material” side:
  - Steel production and Wheat production (including the agriculture land use)





M4: World Economy

Belief

Policy

Data Model

Computes two trajectories
- expected gdp (maxout) without energy constraints
- gdp = constrained growth because of cancellation

GDP is produced from assets that grows according to investments but require energy

Investments are a share of GDP that is reduced two ways
- when energy price goes up, margins reduce
- activities that are cancelled require social management which impairs the ability to invest

State Variables:

- $M_z(y)$: theoretiocal "maxoutput" for zone z = gdp that would have occurred if all necessary energy was here
- $G_z(y)$: gdp for zone z on year y ($G(y) = \Sigma_z G_z(y)$)
- $I_z(y)$: amounts of investments
- $IG_z(y)$: amounts of growth investments in CCEM v8, carbon taxes are re-invested
- $IA_z(y)$: amounts of investments for adaptation
- $SC_z(y)$: steel consummation for zone z at year y
- SP(y): steel price for year y

Parameters

- *margin*(z,p) : impact on profits for remaining activities of zone z when oil-equivalent price is p
- *roiEfficiency*(z,y) : multiplicative factor for return on investment [(R/I) = additional gdp expected R for investment I] in future year y for zone z, based on tech/software and innovation efficiency
- disasterLoss(z,T) : loss of gdp (%) when temperature raises to T
- ironDensity(z,y): density of iron in z economy (gdp / Gt of steel)
- socialExpenses(z,y) : ratio of social costs / GDP in year y

- Alpha(z,t) : fraction of energy that is "redistributed" with subsidy (versus free market)
- CarbonTax(z,y): carbon tax set up in zone z in the year y

- *IRatio*(z) : part of gdp that zone z attributes to investments
- iRevenue(z): share of revenue that is invested
- Energy4steel(y) : energy needed to produce one ton of steel in year y
- DefaultRoI : expected return on inverstment with no constraints and default effiency
- laborFactor(z,y): heuristics for labor competitiveness based on GPD/inhabitant

Equations (functions & differential state)

[1] $M_z(y) = M_z(y - 1) \times (1 - decay) \times HF(y) + IG_z(y - 1) \times roi(z,y)$

$roi(z,y) = DefaultRoI \times roiEfficiency(z,y) \times laborFactor(z,y) \times (1 - socialExpenses(z,y))^2$

[2] $GWD_z(y) = 0.7 \times disasterLoss(z, T(y - 1))$

[3] $G_z(y) = M_z(y) \times (1 - GWD_z(y)) \times (1 - impactCancel(z,y)) \times tradeRatio(z,y)$

[4] $HF(z,y) = (1 - PAIN_z(y) \times productivityFactor) / (1 - PAIN_z(y - 1) \times productivityFactor)$

[5] $impactCancel(z,y) = alpha(z,y) \times (CN_z(y) / (U_z(y) + CN_z(y))) + (1 - alpha(z,y)) \times impact(z,OP_e(y))$

[6] $I_z(y) = G_z(y) \times iRevenue(z) \times (1 - margin(z,oilEquivent(y)) \times (1 - impactCancel(z,p))$

$IG_z(y) = I_z(y) + Tax_z(y) - IA_z(y - 1)$

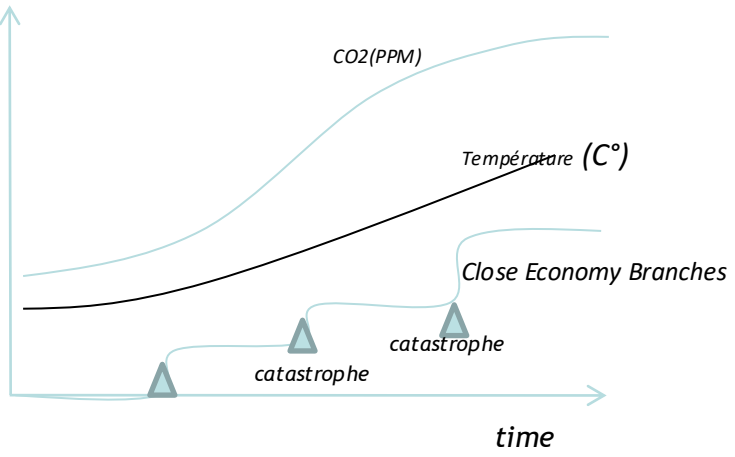
$oilEquivalent(y) = (\Sigma_e P_e(y) \times P_{oil}(1) \times O_e(y) / P_e(1)) / = (\Sigma_c O_e(y))$

[7] $SC_z(y) = G_z(y) / ironDensity(z,y)$

$SP(y) = SP(1) \times (oilEquivalent(y) \times energy4steel(y)) / (oilEquivalent(1)) \times energy4steel(1))$

Ecological Redirection (M5)

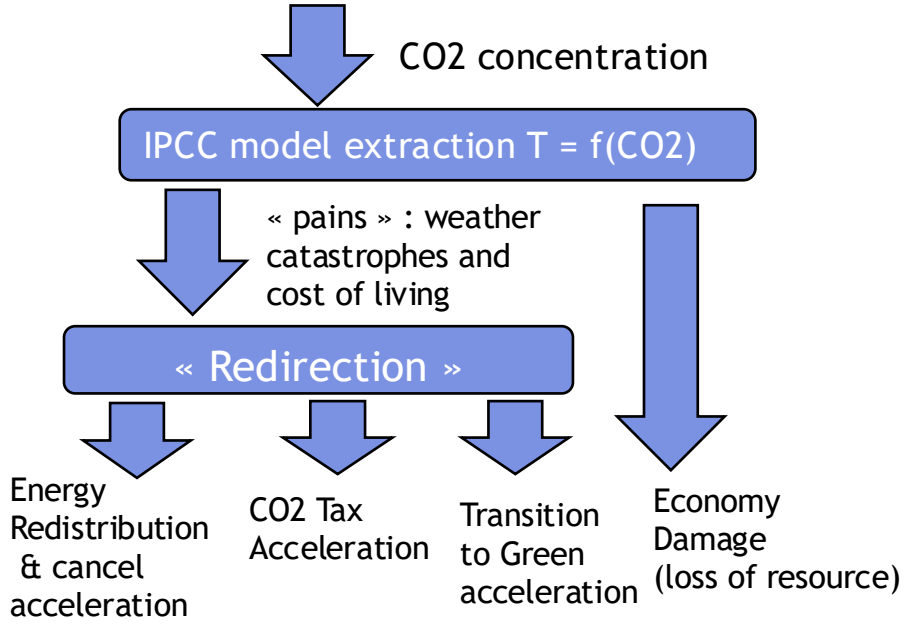
M5 answers the question « ***What kinds of redirection should we expect from the IPCCs global warming consequences*** ?»



Bruno Latour's **redirection** concept tells that this is a non-linear coupling.
There is no "point A to point B" trajectory, but reactions along the way, based on the catastrophic events that will unfold

M5 is made of three components:

- A simple projection from IPCC to link CO2 output (from M2-M3) to expected temperature rise (using RCP 4.5, 6 and 8.5)
- A random, discrete function that define "pain thresholds" as CO2 & temperature rise
- A redirection model (very naïve) with three components
  - Increase into energy redistribution (feedback to M2)
  - Increase CO2 tax (versus planned trajectory)
  - Economic crisis (feedback to M4)
    - ➔ Production capability damaged by warming (e.g. agriculture)
    - ➔ Workforce disruption (from strikes to massive death tolls of natural catastrophes and wars)



Three drivers : fear (of future production loss), pain (less revenue) and compassion (disaster happening elsewhere)

M5: Ecological Redirection

Belief *Policy*

Data Model

We compute multiple impacts of CO2 emissions:

- Average temperature elevation
- Total Area used for agriculture
- Wheat Output (a proxy of agriculture output)
- Reduction of gdp due to global warming
- Pain due to global warming

From the zone level of pain, we compute ecological redirection according to each zone policies, impacting CO2 tax, sobriety (cancel) and energy transition acceleration

Parameters

- *bioHealth*(T,y): percentage of yield evolution (declines when T raises, grows with worldwide diffusion of tech & best practices)
- cropYield(y): yield improvement (as a ratio compared with year 1) expected at year y
- *agroEfficiency*(p) : decline of productivity as energy price increases
- *painProfile*(z) : vector of 3 coefficients that define the global pain level
- painFromClimate(T): step function that sets a pain level as temperature rises

- *pain2<X>*(z,p) : Policy for zone z that links elevated pain level p to a percentage CO2 tax (X=tax), that sets cancel acceleration (sobriety) as a function of pain (X = cancel), sets a target efficiency gain (percent) as a function of pain (X=savings), linear function that links pain level p to Energy Transition acceleration (X=transition), sets the adaptation investment ration (X=Adapt) and sets border taxes against entering trade from CO2 emitting countries (X = *protectionism*)

- *co2Ratio*: additional concentration in the atmosphere from additional CO2 emission (ratio)
- IPCC(c): temperature elevation caused by concentration c, extracted from IPCC RCPs
- *expectedGrowth*(z,dW,dG) : heuristics that defines satisfaction from WheatOutput change and GDP change

State Variables:

- AS(y): Agricultural surface on year y
- ES(y) : Areal that was transferred from Agriculture to Clean Energy Production
- WO(y): Wheat Output
- CO2ppm(y): CO2 concentration reached on year y
- T(y): average globe temperature on year y
- $PAIN_e(y)$: pain factor for zone z at year y
- $TaxF_z(y)$: intensification factor of CO2 tax for z
- $CnF_z(y)$: acceleration of cancel (factor) for zone z
- Delay: number of years between pain and application of policy triggered by pain

Equations (functions & differential state)

[1] $CO2ppm(y) = CO2ppm(y-1) + CO2(y) \times eCO2Ratio$

[2] $T(y) = T(1) + (IPCC(CO2(y)) - IPCC(CO2(1)))$

Using following notation:
$\uparrow F(y) = F(y)/F(1)$

[3] $ES(y) = ES(y-1) + \Delta C_{clean}(y) \times landEImpact(y)$
$AS(y) = (AS(y1) - ES(y)) \times (1 - landLossWarming(T(y)))$
$WO(y) = WO(1) \times \uparrow AS(y) \times \uparrow aggroEfficiency(oilEquivalent(y)) \times cropYield(y) \times bioHealth(y,T(y))$

[4] $PAIN_z(y) = painProfile(z) \times (painFromClimate(T(y)),\ Cn_z(y) \times (1 - Alpha(z),$
$expectedGrowth(z, WO(y) - WO(y-1), (G_z(y) - G_z(y-1)) / Pop_z(y)))$

[5] $TaxF_z(y) = pain2tax(z, PAIN_z(y - Delay))$
$CnF_z(y) = pain2cancel(z, PAIN_z(y - Delay))$
$TrF_z(y) = pain2transition(z, PAIN_z(y - Delay))$
$SvF_z(y) = pain2savings(z, PAIN_z(y - Delay))$
$IAF_z(y) = pain2adapt(z, PAIN_z(y - Delay))$
$PrF_z(y) = pain2protectionism(z, PAIN_z(y - Delay))$

[6] $protect(z1,z2) = \max(0, 1.0 - PrF_{z1}(y) \times$
$\max(0, (CpE(z1,y) - CpE(z2,y)) / 1E\text{-}3 + CpE(z1,y)) \times \max(0, (CpE(z1,y) - CpE(z2,y)) / 1E\text{-}3 + CpE(z1,y))$
$CpE(z,y) = CO2(z,y) / \Sigma_e U_z(e,y))$
$TxR(z,y) = CO2tax(z, PAIN_z(y))$
$IA_z(y) = SvF_z(y - 1) \times G_z(y - 1)$